

OFFLINE SYNTHESIS · training interventions + validation schedules

Aligned static problems

Contacts and rewards fixed
Same boundary $B = (F, X)$
Common feasible conflict: a, b

Full-residual certificates

$\theta_0: L(b) > U(a) + \varepsilon$
 $\theta_1: L(a) > U(b) + \varepsilon$
Overlap / tie / budget: unknown

Strict requirements

$h(\varphi(x^+)) > h(\varphi(x^-))$
Reversal + preservation
Replay all earlier requirements

Exact quotient diagnosis

Cycle $\rightarrow$ no scalar score of φ

$$x_1^+ \rightarrow x_1^- \equiv \varphi \quad x_2^+ \\ x_2^+ \rightarrow x_2^- \equiv \varphi \quad x_1^+$$

LLM: typed catalogue

NodeSet $\rightarrow$ EdgeSet $\rightarrow$ Number

Minimum-cost master

$$\min \sum c_f z_f; \sum d_{Wf} z_f \geq 1$$

Full quotient separation

Surviving cycle: add witness
DAG: observed repair passes
Budget / no feature: unresolved

Concrete arcs + equality joins

cycle: add cut; resolve master

DAG

Freeze ($\varphi, h, \text{kernel}$)

Eligible quality–work tradeoff

LLM rule proposals + joint selection

Replay $s(P)$; assess schedule quality $q(P)$ and compiled work $c(P)$

ONLINE

Current graph $\rightarrow$ compiled $\varphi \rightarrow$ frozen $h \rightarrow$ commit argmax $\rightarrow$ delete closed neighborhood